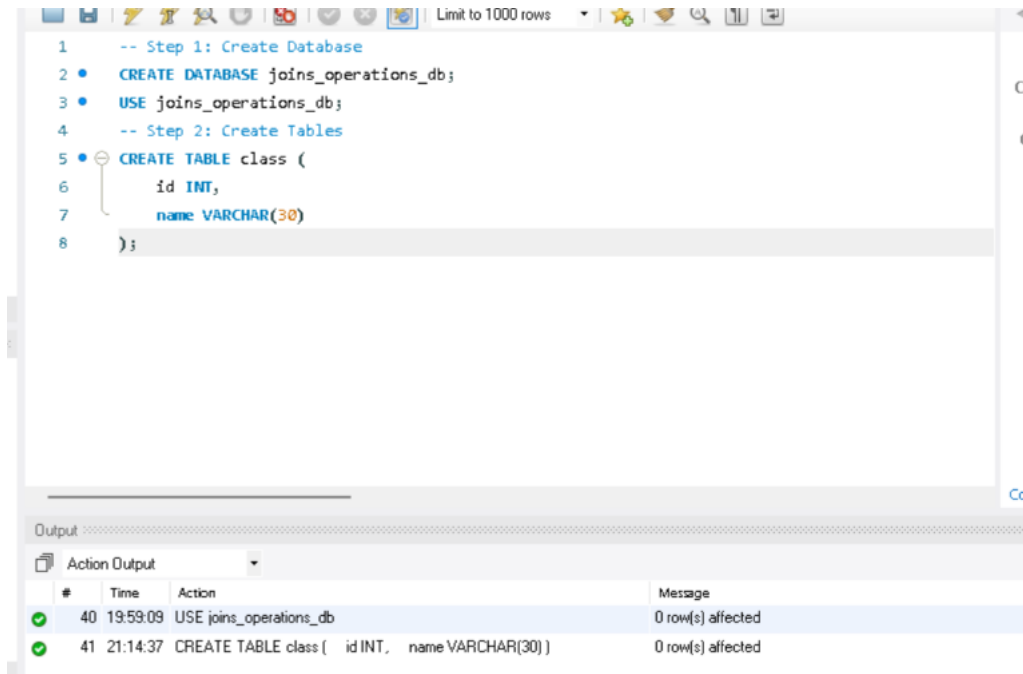


JOINTS

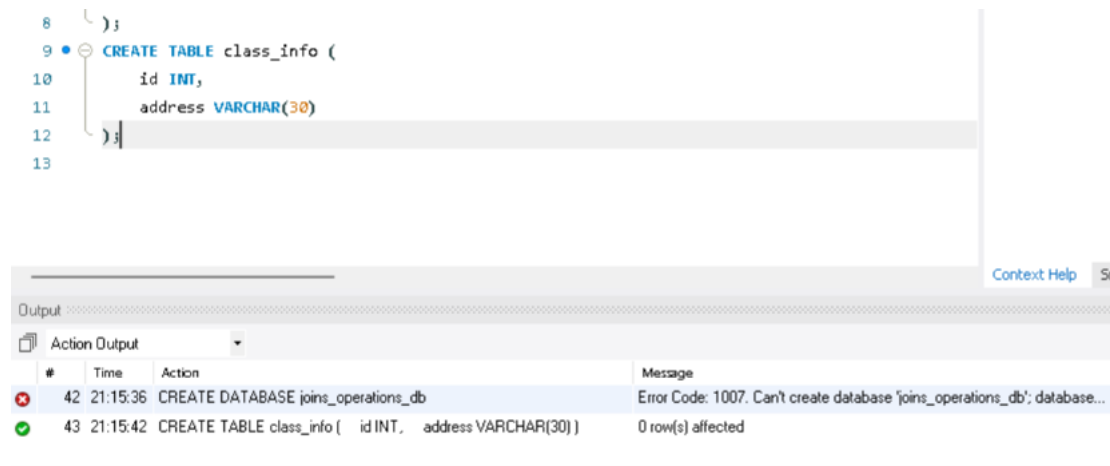
(2520030141 – RAYANKI HIMAVARSHINI)



The screenshot shows a SQL IDE window with a script editor and an output pane. The script contains SQL commands to create a database and a table. The output pane shows two successful actions: creating the database and creating the table.

```
1 -- Step 1: Create Database
2 • CREATE DATABASE joins_operations_db;
3 • USE joins_operations_db;
4 -- Step 2: Create Tables
5 • CREATE TABLE class (
6     id INT,
7     name VARCHAR(30)
8 );
```

#	Time	Action	Message
40	19:59:09	USE joins_operations_db	0 row(s) affected
41	21:14:37	CREATE TABLE class(id INT, name VARCHAR(30))	0 row(s) affected



The screenshot shows a SQL IDE window with a script editor and an output pane. The script contains SQL commands to create a table and a database. The output pane shows an error for creating the database and a successful action for creating the table.

```
8 );
9 • CREATE TABLE class_info (
10     id INT,
11     address VARCHAR(30)
12 );
13
```

#	Time	Action	Message
42	21:15:36	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database...
43	21:15:42	CREATE TABLE class_info(id INT, address VARCHAR(30))	0 row(s) affected

12) ;

13 -- Step 3: Insert Data

14 • INSERT INTO class VALUES

15 (1, 'Hima'),

16 (2, 'Chowdary'),

17 (3, 'Rishika'),

18 (4, 'Narendra'),

19 (5, 'Sujatha');

20

21 • INSERT INTO class_info VALUES

22 (1, 'DELHI'),

23 (2, 'MUMBAI'),

24 (3, 'CHENNAI'),

25 (7, 'NOIDA'),

26 (8, 'PANIPAT');

27

Output

Action Output

#	Time	Action	Message
✓ 44	21:17:39	INSERT INTO class VALUES (1, 'Hima'), (2, 'Chowdary'), (3, 'Rishika'), [...]	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 45	21:17:39	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUMBAI'), (3, 'CHE...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

Auto display current text

Context Help

26 (8, 'PANIPAT');

27 -- Step 4: CROSS JOIN

28 • SELECT *

29 FROM class

30 CROSS JOIN class_info;

31

Result Grid

Filter Rows:

Export: Wrap Cell Content:

	id	name	id	address
▶	5	Sujatha	1	DELHI
	4	Narendra	1	DELHI
	3	Rishika	1	DELHI
	2	Chowdary	1	DELHI
	1	Hima	1	DELHI

Result 2 x

Read Only

Context Help

Snippets

manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 46	21:18:20	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	25 row(s) returned	0.000 sec / 0.000 sec
✓ 47	21:19:04	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	25 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

mysql x

Edit View Query Database Server Tools Scripting Help

SQL Commands SQL File 2" SQL File 3" x

Limit to 1000 rows

9 • CREATE TABLE class_info (
10 id INT,
11 address VARCHAR(30)
12);
13 -- Step 3: Insert Data

Result Grid

	id	name	id	address
3	Rishika	2	MUMBAI	
2	Chowdary	2	MUMBAI	
1	Hima	2	MUMBAI	
5	Sujatha	3	CHENNAI	
4	Naren Sujatha	3	CHENNAI	
3	Rishika	3	CHENNAI	
2	Chowdary	3	CHENNAI	
1	Hima	3	CHENNAI	
5	Sujatha	7	NOIDA	
4	Narendra	7	NOIDA	
3	Rishika	7	NOIDA	
2	Chowdary	7	NOIDA	
1	Hima	7	NOIDA	
5	Sujatha	8	PANIPAT	
4	Narendra	8	PANIPAT	
3	Rishika	8	PANIPAT	

Result 2 x

Read Only Context Help Snippets

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Limit to 1000 rows

28 • SELECT *
29 FROM class
30 CROSS JOIN class_info;
31 -- Step 5: INNER JOIN
32 • SELECT *
33 FROM class
34 INNER JOIN class_info
35 ON class.id = class_info.id;
36

Result Grid

	id	name	id	address
1	Hima	1	DELHI	
2	Chowdary	2	MUMBAI	
3	Rishika	3	CHENNAI	

Result 3 x

Read Only Context

Output

Action Output

#	Time	Action	Message
47		Select output pane	
48	21:20:45	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	25 row(s) returned
48	21:20:45	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id	3 row(s) returned

SQL Commands SQL File 2" SQL File 3" SQL Additions

Limit to 1000 rows

```

32 • SELECT *
33 FROM class
34 INNER JOIN class_info
35 ON class.id = class_info.id;
36 -- Step 6: INNER JOIN Selected Columns
37 • SELECT class.name, class_info.address
38 FROM class
39 INNER JOIN class_info
40 ON class.id = class_info.id;

```

Result Grid

	name	address
▶	Hima	DELHI
	Chowdary	MUMBAI
	Rishika	CHENNAI

Result 4

Output

Action Output

#	Time	Action	Message
✖	49 21:22:06	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database...
✔	50 21:22:09	SELECT class.name, class_info.address FROM class INNER JOIN class...	3 row(s) returned

Automatic disabled. Use manually current can toggle a

```

36 -- Step 6: INNER JOIN Selected Columns
37 • SELECT class.name, class_info.address
38 FROM class
39 INNER JOIN class_info
40 ON class.id = class_info.id;
41 -- Step 7: NATURAL JOIN
42 • SELECT *
43 FROM class
44 NATURAL JOIN class_info;

```

Result Grid

	id	name	address
▶	1	Hima	DELHI
	2	Chowdary	MUMBAI
	3	Rishika	CHENNAI

Result 5

Output

Action Output

#	Time	Action	Message
✖	51 21:22:42	CREATE DATABASE joins_operations_db	Error Code: 1007. Can't create database 'joins_operations_db'; database...
✔	52 21:22:44	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned

Automatic disabled. Use manually current can toggle a

```

42 • SELECT *
43 FROM class
44 NATURAL JOIN class_info;
45 -- Step 8: LEFT OUTER JOIN
46 • SELECT *
47 FROM class
48 LEFT OUTER JOIN class_info
49 ON class.id = class_info.id;
--

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	id	name	id	address
▶	1	Hima	1	DELHI
	2	Chowdary	2	MUMBAI
	3	Rishika	3	CHENNAI
	4	Narendra	NULL	NULL
	5	Sujatha	NULL	NULL

Result Grid

Result 6 x

Read Only

Output

Action Output

	#	Time	Action	Message
✓	52	21:22:44	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
✓	53	21:23:28	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = cla...	5 row(s) returned

```

48 LEFT OUTER JOIN class_info
49 ON class.id = class_info.id;
50 -- Step 9: LEFT JOIN with NULL
51 • SELECT *
52 FROM class
53 LEFT JOIN class_info
54 ON class.id = class_info.id
55 WHERE class_info.id IS NULL;
--

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	id	name	id	address
▶	4	Narendra	NULL	NULL
	5	Sujatha	NULL	NULL

Result 7 x

Re

Output

Action Output

	#	Time	Action	Message
✓	53	21:23:28	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = cla...	5 row(s) returned
✓	54	21:24:05	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id...	2 row(s) returned

```

54     ON class.id = class_info.id
55     WHERE class_info.id IS NULL;
56     -- Step 10: RIGHT OUTER JOIN
57 •   SELECT *
58     FROM class
59     RIGHT OUTER JOIN class_info
60     ON class.id = class_info.id;
61
--

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	id	name	id	address
▶	1	Hima	1	DELHI
	2	Chowdary	2	MUMBAI
	3	Rishika	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 8 Read Only

Output

Action Output

#	Time	Action	Message
✓	54 21:24:05	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id...	2 row(s) returned
✓	55 21:24:40	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = cl...	5 row(s) returned

SQL EDITORS SQL FILE 2 SQL FILE 3

Limit to 1000 rows

```

60     ON class.id = class_info.id;
61     -- Step 11: RIGHT JOIN with NULL
62 •   SELECT *
63     FROM class
64     RIGHT JOIN class_info
65     ON class.id = class_info.id
66     WHERE class.id IS NULL;
67
68
--

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	id	name	id	address
▶	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 9 Read Only

Output

Action Output

#	Time	Action	Message
✓	55 21:24:40	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = cl...	5 row(s) returned
✓	56 21:25:19	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info...	2 row(s) returned

```

69 • SELECT *
70 FROM class
71 LEFT JOIN class_info
72 ON class.id = class_info.id
73 UNION
74 SELECT *
75 FROM class
76 RIGHT JOIN class_info
77 ON class.id = class_info.id;
--

```

Automatically disable manual current toggle

Result Grid

	id	name	id	address
▶	1	Hima	1	DELHI
	2	Chowdary	2	MUMBAI
	3	Rishika	3	CHENNAI
	4	Narendra	NULL	NULL
	5	Sujatha	NULL	NULL

Result 10

Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 56	21:25:19	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id...	2 row(s) returned
✓ 57	21:26:10	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id...	7 row(s) returned

```

81 LEFT JOIN class_info
82 ON class.id = class_info.id
83 WHERE class_info.id IS NULL
84 UNION
85 SELECT *
86 FROM class
87 RIGHT JOIN class_info
88 ON class.id = class_info.id
89 WHERE class.id IS NULL;
--

```

Limit to 1000 rows

Au disa me cun b

Result Grid

	id	name	id	address
▶	4	Narendra	NULL	NULL
	5	Sujatha	NULL	NULL
	7	NOIDA	NULL	NULL
	8	PANIPAT	NULL	NULL

Result 11

Read Only Context

Output

Action Output

#	Time	Action	Message
✓ 57	21:26:10	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id...	7 row(s) returned
✓ 58	21:27:21	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id...	4 row(s) returned

```

88     ON class.id = class_info.id
89     WHERE class.id IS NULL;
90     -- Step 14: Create Set Operation Tables
91     • CREATE TABLE first_table (
92         id INT,
93         name VARCHAR(30)
94     );
95     • CREATE TABLE second_table (
96         id INT,
97         name VARCHAR(30)
98     );
99
100

```

Output

Action Output

#	Time	Action	Message
✓ 59	21:28:30	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
✓ 60	21:28:30	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected

```

99     -- Step 15: Insert Set Operation Data
100    • INSERT INTO first_table VALUES
101        (1, 'Hima'),
102        (2, 'Sujatha');
103    • INSERT INTO second_table VALUES
104        (2, 'Narendra'),
105        (3, 'Rishika');
106
107
108
109

```

Output

Action Output

#	Time	Action	Message
✓ 61	21:29:53	INSERT INTO first_table VALUES (1, 'Hima'), (2, 'Sujatha')	2 row(s) affected Records: 2 Duplicates: 0
✓ 62	21:29:53	INSERT INTO second_table VALUES (2, 'Narendra'), (3, 'Rishika')	2 row(s) affected Records: 2 Duplicates: 0

```

106    -- Step 16: UNION
107    • SELECT *
108    FROM first_table
109    UNION
110    SELECT *
111    FROM second_table;

```

Result Grid

	id	name
1	1	Hima
2	2	Sujatha
	2	Narendra
3	3	Rishika

Result 12 x

Output

Action Output

#	Time	Action	Message	Duration
✓ 62	21:29:53	INSERT INTO second_table VALUES (2, 'Narendra'), (3, 'Rishika')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.00s
✓ 63	21:30:36	SELECT * FROM first_table UNION SELECT * FROM second_table	4 row(s) returned	0.00s

manually get
current caret
toggle auto


```

111 FROM second_table;
112 -- Step 17: UNION Names
113 • SELECT name
114 FROM first_table
115 UNION
116 SELECT name
117 FROM second_table;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	name
▶	Hima
	Sujatha
	Narendra
	Rishika

Result 13

Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 63	21:30:36	SELECT * FROM first_table UNION SELECT * FROM second_table	4 row(s) returned
✓ 64	21:31:27	SELECT name FROM first_table UNION SELECT name FROM second_...	4 row(s) returned

disap
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to

```

117 FROM second_table;
118 -- Step 18: UNION ALL
119 • SELECT *
120 FROM first_table
121 UNION ALL
122 SELECT *
123 FROM second_table;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name
▶	1	Hima
	2	Sujatha
	2	Narendra
	3	Rishika

Result 14

Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 64	21:31:27	SELECT name FROM first_table UNION SELECT name FROM second_...	4 row(s) returned
✓ 65	21:32:20	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned

manual
current
toggle

SQL Commands SQL File 2* SQL File 3* SQL Additions

Limit to 1000 rows

```

123 FROM second_table;
124 -- Step 19: Count UNION ALL Records
125 • SELECT COUNT(*)
126 FROM (
127     SELECT * FROM first_table
128     UNION ALL
129     SELECT * FROM second_table
130 ) AS A;
131

```

Result Grid

COUNT(*)
4

Result 15

Output

Action Output

#	Time	Action	Message	Duration
65	21:32:20	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000
66	21:33:09	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) AS A	1 row(s) returned	0.000

Automatic commit disabled. Use manually get current caret position, toggle auto commit.

SQL Commands SQL File 2* SQL File 3* SQL Additions

Limit to 1000 rows

```

129 SELECT * FROM second_table;
130 ) AS A;
131 -- Step 20: INTERSECT
132 • SELECT *
133 FROM first_table
134 INTERSECT
135 SELECT *
136 FROM second_table;
137

```

Result Grid

id	name
----	------

Result 16

Output

Action Output

#	Time	Action	Message	Duration
66	21:33:09	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) AS A	1 row(s) returned	0.000
67	21:35:13	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	0 row(s) returned	0.000

Automatic commit disabled. Use manually get current caret position, toggle auto commit.

```

136 FROM second_table;
137 -- Step 21: INTERSECT Names
138 • SELECT name
139 FROM first_table
140 INTERSECT
141 SELECT name
142 FROM second_table;
143

```

Automatic context disabled. Use the toggle button to manually get help at the current caret position.

Result Grid

name

Result 17 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
68	21:35:57	- Step 21: INTERSECT Names SELECT name FROM first_table INTER...	Error Code: 1064. You have an error in your SQL syntax; check the man...
69	21:36:07	SELECT name FROM first_table INTERSECT SELECT name FROM se...	0 row(s) returned

SQL Commands SQL File 2" SQL File 3" x

```

145 • SELECT f.*
146 FROM first_table f
147 WHERE NOT EXISTS (
148     SELECT 1
149     FROM second_table s
150     WHERE s.id = f.id
151     AND s.name = f.name
152 );
153

```

Automatic context disabled. Use the toggle button to manually get help at the current caret position.

Result Grid

id	name
1	Hima
2	Sujatha

first_table 18 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Ret
69	21:36:07	SELECT name FROM first_table INTERSECT SELECT name FROM se...	0 row(s) returned	0.000 sec / 0
70	21:37:16	SELECT f.* FROM first_table f WHERE NOT EXISTS (SELECT 1 ...	2 row(s) returned	0.015 sec / 0

SQL CommandsSQL File 2"SQL File 3"SQL Additions

Limit to 1000 rows

Automatic context disabled. Use the manually get help current caret position toggle automatically

```
152 };  
153 -- Step 23: MINUS Names  
154 • SELECT f.name  
155 FROM first_table f  
156 WHERE NOT EXISTS (  
157     SELECT 1  
158     FROM second_table s  
159     WHERE s.name = f.name  
160 );
```

Result Grid

Filter Rows: Export: Wrap Cell Content: Result Grid

name
Hima
Sujatha

first_table 19 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fe
70	21:37:16	SELECT f.* FROM first_table f WHERE NOT EXISTS (SELECT 1 ...	2 row(s) returned	0.015 sec / I
71	21:38:01	SELECT f.name FROM first_table f WHERE NOT EXISTS (SELECT ...	2 row(s) returned	0.000 sec / I